7.8.3 Global and local variables

Here, we will show how to use global variables and local variables in the same code.

Example : Using Global and Local variables in same code

```
x=8                    # x is a global variable
def loc():
        global x
        y = "local"
        x = x * 2
        print(x)
        print(y)
loc()
```

Output:

```
16
local
```

In the above program, we declare ***x*** as global and ***y*** as local variable in the function **loc()**.

After calling the function **loc()**, the value of ***x*** becomes 16 because we used **x=x * 2.** After that, we print the value of local variable ***y*** i.e. local.

Example : Global variable and Local variable with same name

```
x = 5
def loc():
        x = 10
        print ("local x:", x)
loc()
print ("global x:", x)
```

Output:

```
local x: 10
global x: 5
```

In above code, we used same name 'x' for both global variable and local variable. We get different result when we print same variable because the variable is declared in both scopes, i.e. the local scope inside the function loc() and global scope outside the function loc().

The output :- local x: 10, is called local scope of variable.

The output:- global x: 5, is called global scope of variable.

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